

SIMATS ENGINEERING

ASSESSMENT TOOL 3 : Technical Presentation

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions. (BL2)*

Assessment Tool Weightage: 10%

Dave's Taxonomy: Imitation (Level 1)
Question Count: 5

Total Marks: 50

Code of Conduct

I S.DIVYASRI(192524388)_____ (Name / Reg. No.) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: __S.DIVYASRI_____

Assessment Tasks

Technical Presentation Topics

Prepare a 6-8 minute presentation on any one of the following topics. Your slides must include definitions, architecture, industry relevance, and one practical example.

Here are 10 additional technical presentation topics in a similar style, suitable for a 6–8 minute presentation with definitions, architecture, industry relevance, and a practical example:

1. **Containerization and Cloud-Native Applications: Transforming Modern Software Deployment**
 - Explore containers, container architecture, benefits, and a real-world deployment example.
2. **Serverless Computing: Reducing Infrastructure Management in the Cloud**
 - Discuss Function-as-a-Service (FaaS), architecture, business advantages, and practical use cases.
3. **Edge Computing vs Cloud Computing: Choosing the Right Computing Paradigm**
 - Compare architectures, applications, and industry adoption scenarios.
4. **Microservices Architecture in Cloud Environments**
 - Explain microservices design, benefits over monolithic systems, and industry implementations.
5. **Cloud Security Fundamentals: Protecting Data and Applications**
 - Cover security models, shared responsibility architecture, and real-world security practices.
6. **Multi-Cloud and Hybrid Cloud Strategies for Enterprise IT**
 - Analyze deployment models, business benefits, challenges, and practical examples.
7. **The Role of Artificial Intelligence and Machine Learning in Cloud Platforms**
 - Discuss AI/ML cloud services, architecture, and industry applications.
8. **DevOps and Continuous Integration/Continuous Deployment (CI/CD) in Cloud Computing**

- Explain DevOps principles, CI/CD architecture, and practical implementation examples.
- 9. Cloud Storage Technologies: Object, Block, and File Storage Compared**
 - Compare storage architectures, use cases, and industry adoption patterns.
- 10. Internet of Things (IoT) and Cloud Computing: Building Connected Ecosystems**
 - Examine IoT-cloud architecture, benefits, challenges, and real-world applications.
- 11. Cloud Migration Strategies: Moving Enterprise Applications to the Cloud**
 - Examine migration approaches, cloud architecture considerations, benefits, challenges, and a successful migration case study.
- 12. Big Data Analytics in Cloud Computing**
 - Explore cloud-based data processing architectures, analytics platforms, business value, and real-world applications.
- 13. Identity and Access Management (IAM) in Cloud Environments**
 - Discuss authentication, authorization architectures, security benefits, and practical enterprise implementations.
- 14. Content Delivery Networks (CDNs) and Their Role in Cloud Services**
 - Explain CDN architecture, performance optimization, industry relevance, and examples from popular web platforms.

Minimum slide flow:

- Title and objective
- Core technical explanation
- Architecture or process diagram
- Real-world use case
- Conclusion and key takeaway
- Limitations and Future Scope

Technical Presentation Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Technical Content	30%	Content is highly accurate, relevant, and insightful	Content is correct with minor gaps	Basic content with limited depth	Content is inaccurate or superficial
Structure and Flow	20%	Excellent logical flow and slide organization	Good structure with minor issues	Some organization but weak flow	Disorganized presentation
Use of Visuals	15%	Visuals strongly support technical communication	Relevant visuals used well	Limited or unclear visuals	No meaningful visuals
Communication Skills	20%	Confident, clear, and engaging delivery	Clear delivery with minor hesitation	Moderate clarity but uneven delivery	Weak delivery and low clarity
Professionalism	15%	Well-prepared and audience-focused presentation	Mostly professional	Basic preparation visible	Poor preparation and professionalism

CSA1707-CLOUD COMPUTING AND BIG DATA ANALYSIS



The Role of Artificial Intelligence and Machine Learning in Cloud Platforms

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GUIDED BY: DR.J.CHENNI KUMARAN

OBJECTIVE & AGENDA

Why AI and ML Matter in the Cloud

OBJECTIVE

To examine how cloud platforms deliver Artificial Intelligence and Machine Learning as scalable, on-demand services — and how this convergence is transforming modern software and business systems.

3 Major cloud AI/ML service categories

24/7 Elastic, on-demand model availability

Zero Infrastructure to manage in-house

1 Core technical explanation of AI/ML in the cloud

2 Cloud AI/ML architecture & service stack

3 End-to-end ML pipeline (process diagram)

4 Industry applications across sectors

5 Real-world enterprise use case

6 Limitations, challenges & future scope

CORE TECHNICAL EXPLANATION

What is AI/ML in Cloud Computing?

Cloud computing solves this problem by providing AI and ML as online services. Companies can access these resources whenever they need them and pay only for what they use."

AIaaS

AI-as-a-Service

AIaaS provides ready-made AI models through APIs. This means developers don't need to create AI models from scratch. They simply connect their application to the cloud service..

MLaaS

ML-as-a-Service

Unlike AIaaS, MLaaS is designed for developers and data scientists who want to build their own machine learning models. It provides tools for collecting data, training models, testing them, and deploying them without worrying about hardware.

KEY CLOUD BUILDING BLOCKS

GPU

Elastic Compute

GPU / TPU clusters that scale on demand for training & inference

DB

Data & Storage

Object storage and data lakes feeding model training pipelines

ML

Managed ML Platforms

SageMaker, Vertex AI, Azure ML for build-train-deploy workflows

API

Pre-built AI APIs

Vision, speech, NLP and translation services ready to integrate

The Cloud AI/ML Service Stack

APPLICATION LAYER

End-user apps: chatbots, recommendation engines, fraud alerts, forecasting dashboards

AI SERVICE LAYER

Pre-built APIs — Vision, Speech-to-Text, NLP, Translation, Recommendation engines

ML PLATFORM LAYER

MLOps tooling — model training, versioning, registry, deployment & monitoring

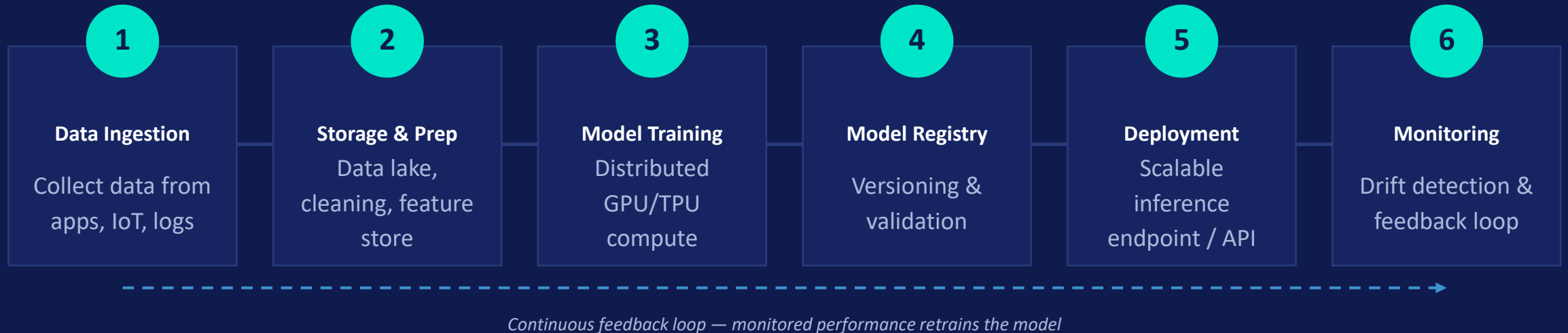
INFRASTRUCTURE LAYER

Elastic compute (GPU/TPU), distributed storage, high-speed networking

*Higher abstraction
(less management overhead)*

*Lower abstraction
(more control & customization)*

End-to-End ML Pipeline on the Cloud



Where Cloud AI/ML Creates Impact



Healthcare

Diagnostic imaging, patient risk prediction, drug discovery acceleration



Finance

Real-time fraud detection, algorithmic trading, credit risk scoring



Retail

Personalized recommendations, demand forecasting, dynamic pricing



Manufacturing

Predictive maintenance, defect detection, supply-chain optimization



Education

Adaptive learning platforms, automated grading, plagiarism detection



Transportation

Route optimization, autonomous systems, predictive fleet maintenance

Netflix — Personalization at Global Scale

THE CHALLENGE

Deliver personalized content recommendations to 300M+ subscribers worldwide, in real time, without over-provisioning infrastructure for peak demand.

THE APPROACH

Netflix runs its recommendation and encoding ML models on AWS's elastic cloud infrastructure, using auto-scaling compute and managed data pipelines to retrain models continuously on viewing behavior.

MEASURABLE OUTCOMES

80%+

of content watched originates from AI-driven recommendations

1M+

GPU/CPU compute hours flexibly scaled via the cloud each year

\$1B

estimated annual value from reduced churn via personalization

Illustrative figures compiled from publicly reported industry commentary on Netflix's cloud-based personalization system.

Challenges Today, Directions Ahead

LIMITATIONS & CHALLENGES

- **Cost unpredictability**
GPU/TPU usage and data egress fees can escalate quickly at scale
- **Data privacy & compliance**
Sensitive data in shared cloud environments raises regulatory risk
- **Vendor lock-in**
Proprietary ML APIs and formats limit portability across providers
- **Latency constraints**
Real-time inference can suffer from network round-trip delays
- **Model bias & explainability**
Black-box cloud models complicate fairness and auditability

FUTURE SCOPE

- **Edge AI integration**
Inference moves closer to devices, cutting latency and bandwidth
- **Federated learning**
Models train across distributed data without centralizing it
- **AutoML & no-code tools**
Democratizing model building for non-specialist users
- **Generative AI services**
Foundation models offered as flexible, fine-tunable cloud APIs
- **Greener, efficient compute**
Specialized AI chips reducing the energy cost of cloud ML

CONCLUSION

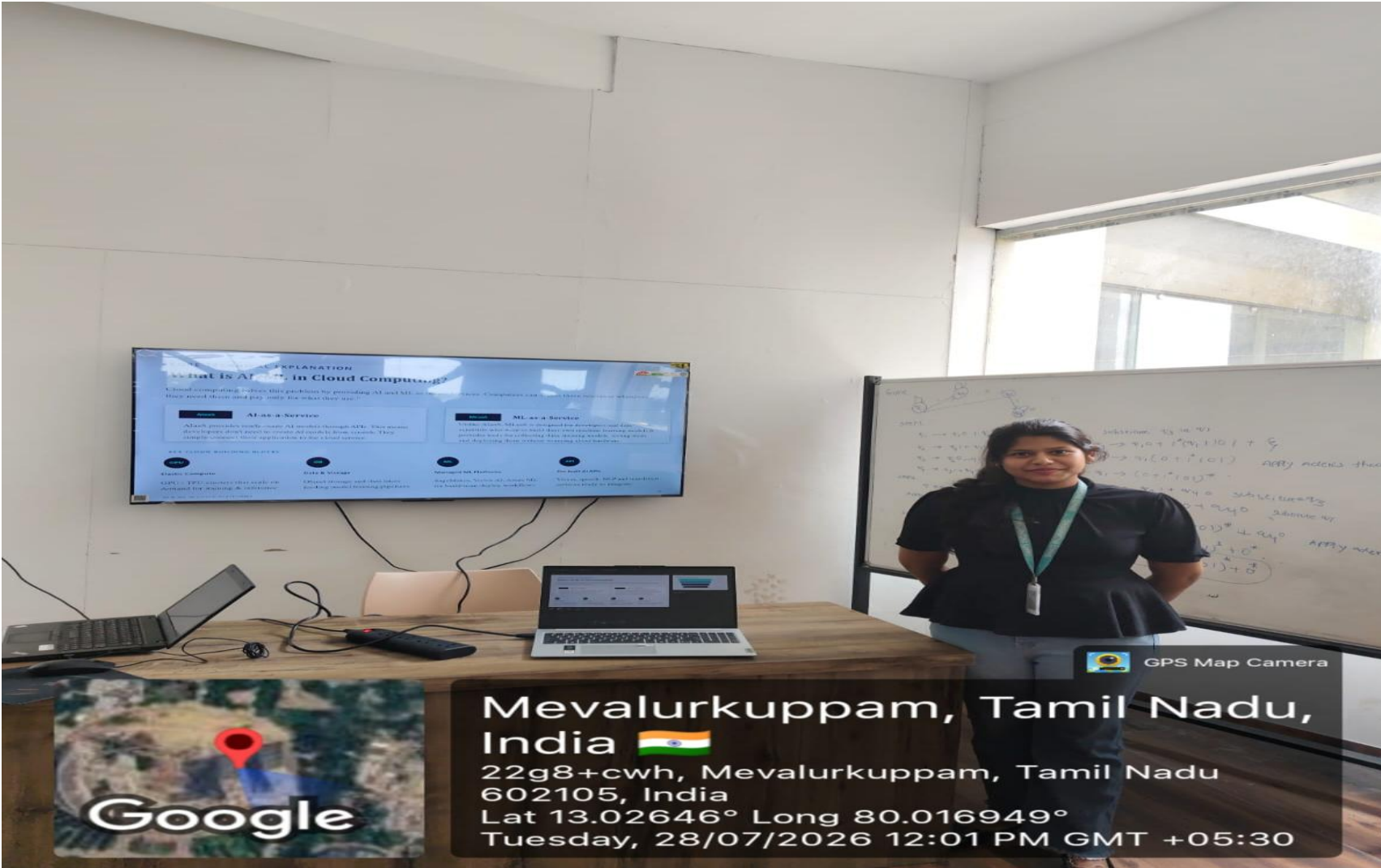
Cloud is the Engine Behind Applied AI

By pairing elastic compute with managed ML platforms, cloud providers have turned artificial intelligence from a specialist research capability into an accessible, on-demand utility for organizations of every size.

- ✓ AI/ML in the cloud reduces cost, time, and expertise barriers to adoption
- ✓ A layered architecture — infrastructure, platform, services — enables flexibility
- ✓ Real impact already spans healthcare, finance, retail, and manufacturing
- ✓ Success requires actively managing cost, privacy, and vendor-lock-in risks

KEY TAKEAWAY

Cloud-delivered AI/ML is not a passing trend — it is now foundational infrastructure for building intelligent, scalable, real-world applications.



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